

SMAPhysics.com

JEE Main Physics Practice Sheet — DPP-040

Topic: Power — Definition, Average Power, Instantaneous Power & Constant Power Motion

Time Allowed: 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

- Q1.** A body of mass m accelerates from rest under the action of an engine delivering constant power P . The velocity v of the body as a function of time t is proportional to:
- (A) t
(B) $t^{1/2}$
(C) $t^{3/2}$
(D) t^2
- Q2.** In Question 1, the displacement s of the body from rest as a function of time t is proportional to:
- (A) $t^{3/2}$
(B) $t^{1/2}$
(C) t^2
(D) $t^{5/2}$
- Q3.** An electric motor delivers $P = 2.0$ kW of power to lift a water tank of mass $M = 300$ kg to a height of $h = 20$ m. If the efficiency of the motor is 75%, the time taken to fill the tank is ($g = 10$ m/s²):
- (A) 30 s
(B) 45 s
(C) 40 s
(D) 60 s
- Q4.** A constant force $\vec{F} = (4\hat{i} + 3\hat{j} - 2\hat{k})$ N acts on a particle moving with an instantaneous velocity $\vec{v} = (2\hat{i} - 4\hat{j} + 5\hat{k})$ m/s. The instantaneous power delivered by the force is:
- (A) -14 W
(B) +14 W
(C) -6 W
(D) +26 W
- Q5.** A body of mass $m = 2$ kg is projected vertically upwards with an initial speed $u = 20$ m/s. The instantaneous power delivered by the gravitational force at time $t = 1$ s is ($g = 10$ m/s²):
- (A) -200 W
(B) +200 W
(C) -400 W
(D) Zero
- Q6.** An engine pumps water continuously through a pipe of cross-sectional area A . If water emerges with a speed v , and the density of water is ρ , the rate at which work is done by the engine to maintain this flow is:
- (A) $\frac{1}{2}\rho Av^2$
(B) $\frac{1}{2}\rho Av^3$
(C) ρAv^3
(D) $\frac{1}{4}\rho Av^3$
- Q7.** To increase the flow velocity of water in Question 6 by a factor of 2 ($v' = 2v$), the power delivered by the engine must be increased by a factor of:
- (A) 2
(B) 4
(C) 8
(D) 16
- Q8.** An automobile of mass m accelerates uniformly from rest to speed v_0 in time t_0 . The instantaneous power delivered by the engine at time t ($t \leq t_0$) is:
- (A) $\frac{mv_0^2}{t_0^2}t$
(B) $\frac{mv_0^2}{2t_0^2}t$
(C) $\frac{mv_0^2}{t_0}$
(D) $\frac{mv_0}{t_0^2}t^2$
- Q9.** In Question 8, the average power delivered by the engine over the entire interval from $t = 0$ to $t = t_0$ is:
- (A) $\frac{mv_0^2}{t_0}$
(B) $\frac{mv_0^2}{2t_0}$
(C) $\frac{mv_0^2}{4t_0}$
(D) $\frac{2mv_0^2}{t_0}$
- Q10.** A car of mass $M = 1000$ kg climbs an incline of 1 in 20 ($\sin \theta = 0.05$) at a constant speed of $v = 15$ m/s. If the resistance to motion due to friction is $f = 200$ N, the power developed by the engine is ($g = 10$ m/s²):
- (A) 7.5 kW
(B) 3.0 kW
(C) 10.5 kW
(D) 15.0 kW
- Q11.** A particle of mass m moves in a straight line such that its displacement x depends on time t as $x = \alpha t^3 + \beta t$. The instantaneous power delivered to the particle is proportional to:
- (A) t
(B) t^3
(C) $(3\alpha t^2 + \beta)t$

(D) t^2

Q12. A vehicle of mass m starts from rest and moves with constant power P . The velocity v acquired after travelling a distance s is:

(A) $\left(\frac{3Ps}{m}\right)^{1/3}$

(B) $\left(\frac{2Ps}{m}\right)^{1/2}$

(C) $\left(\frac{Ps}{m}\right)^{1/3}$

(D) $\left(\frac{3Ps}{2m}\right)^{1/3}$

Q13. A particle moves along the x -axis from rest at $x = 0$ under a constant power source. The acceleration a of the particle as a function of its displacement x is proportional to:

(A) $x^{-1/3}$

(B) $x^{-1/2}$

(C) $x^{-2/3}$

(D) x^{-1}

Q14. A crane lifts a load of mass $M = 500$ kg vertically upward with a constant acceleration $a = 2$ m/s². The instantaneous power developed by the crane motor at $t = 3$ s after starting from rest is ($g = 10$ m/s²):

(A) 18 kW

(B) 36 kW

(C) 30 kW

(D) 15 kW

Q15. An elevator cabin of mass $M = 1800$ kg is moving upward at a constant speed $v = 3$ m/s against a constant frictional resistance force $f = 4000$ N. The minimum horsepower (hp) required from the motor is (1 hp = 746 W, $g = 10$ m/s²):

(A) 88.5 hp

(B) 29.5 hp

(C) 59.0 hp

(D) 72.4 hp

Q16. A machine gun fires N bullets per second, each of mass m with muzzle speed u . The power delivered by the gun mechanism to fire the bullets is:

(A) $\frac{1}{2}Nmu^2$

(B) Nmu^2

(C) $\frac{1}{2}N^2mu^2$

(D) Nmu

Q17. The power supplied to a particle of mass $m = 2$ kg varies with time as $P(t) = 3t^2$ W. If the particle was initially at rest at $t = 0$, its velocity at $t = 2$ s is:

(A) $2\sqrt{2}$ m/s

(B) 4.0 m/s

(C) 2.0 m/s

(D) 8.0 m/s

Q18. A wind turbine with blades sweeping an area A is placed in a wind stream of speed v . If the air density is ρ , the maximum theoretical power of the incoming wind stream is:

(A) $\frac{1}{2}\rho Av^2$

(B) ρAv^3

(C) $\frac{1}{2}\rho Av^3$

(D) $\frac{1}{4}\rho Av^3$

Q19. A particle of mass m is moving in a circle of radius R with uniform speed v . The instantaneous power delivered by the net centripetal force is:

(A) $\frac{mv^3}{R}$

(B) $\frac{mv^2}{R}$

(C) $\frac{1}{2}\frac{mv^3}{R}$

(D) Zero

Q20. A car travels with constant speed v along a horizontal road where resistive aerodynamic drag is $F_{\text{drag}} = cv^2$. The engine power output required to maintain this speed is proportional to:

(A) v^2

(B) v^3

(C) v^4

(D) v

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-040)

Power — Definition, Average & Instantaneous Power, Constant Power Motion

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	C	16	A
2	A	7	C	12	A	17	A
3	C	8	A	13	A	18	C
4	A	9	B	14	B	19	D
5	A	10	C	15	A	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (B)

Constant power: $P = Fv = \left(m \frac{dv}{dt}\right) v$.

$$v dv = \frac{P}{m} dt \Rightarrow \int_0^v v dv = \frac{P}{m} \int_0^t dt$$

$$\frac{v^2}{2} = \frac{P}{m} t \Rightarrow v = \sqrt{\frac{2P}{m}} t^{1/2} \propto t^{1/2}$$

Sol 2. Option (A)

From Solution 1: $v = \frac{ds}{dt} = \sqrt{\frac{2P}{m}} t^{1/2}$.

$$s = \int_0^t \sqrt{\frac{2P}{m}} t^{1/2} dt = \sqrt{\frac{2P}{m}} \left(\frac{t^{3/2}}{3/2}\right) \propto t^{3/2}$$

Sol 3. Option (C)

Useful work required:

$$W_{\text{useful}} = Mgh = 300 \times 10 \times 20 = 60000 \text{ J}$$

Effective power available:

$$P_{\text{eff}} = \eta P_{\text{input}} = 0.75 \times 2000 = 1500 \text{ W}$$

Time taken:

$$t = \frac{W_{\text{useful}}}{P_{\text{eff}}} = \frac{60000}{1500} = 40 \text{ s}$$

Sol 4. Option (A)

Instantaneous power: $P = \vec{F} \cdot \vec{v}$.

$$P = (4)(2) + (3)(-4) + (-2)(5) = 8 - 12 - 10 = -14 \text{ W}$$

Sol 5. Option (A)

Velocity at $t = 1 \text{ s}$:

$$v = u - gt = 20 - 10(1) = 10 \text{ m/s (upward)}$$

Gravitational force:

$$F_g = mg = 2 \times 10 = 20 \text{ N (downward)}$$

$$P = \vec{F}_g \cdot \vec{v} = -(20)(10) = -200 \text{ W}$$

Sol 6. Option (B)

Mass of water flowing per unit time: $\frac{dm}{dt} = \rho Av$.

$$P = \frac{dK}{dt} = \frac{1}{2} \left(\frac{dm}{dt}\right) v^2 = \frac{1}{2} (\rho Av) v^2 = \frac{1}{2} \rho Av^3$$

Sol 7. Option (C)

Since $P \propto v^3$:

$$\frac{P'}{P} = \left(\frac{2v}{v}\right)^3 = 2^3 = 8$$

Sol 8. Option (A)

Acceleration: $a = \frac{v_0}{t_0}$.

Force: $F = ma = \frac{mv_0}{t_0}$.

Velocity at time t : $v(t) = at = \left(\frac{v_0}{t_0}\right) t$.

Instantaneous power:

$$P(t) = Fv(t) = \left(\frac{mv_0}{t_0}\right) \left(\frac{v_0}{t_0} t\right) = \frac{mv_0^2}{t_0^2} t$$

Sol 9. Option (B)

Total work done: $W = \Delta K = \frac{1}{2} mv_0^2$.

$$P_{\text{avg}} = \frac{W}{t_0} = \frac{mv_0^2}{2t_0}$$

Sol 10. Option (C)

Total opposing force:

$$F_{\text{total}} = Mg \sin \theta + f = 1000(10)(0.05) + 200 = 500 + 200 = 700 \text{ N}$$

Power required at constant velocity:

$$P = F_{\text{total}} v = 700 \times 15 = 10500 \text{ W} = 10.5 \text{ kW}$$

Sol 11. Option (C)

$$v = \frac{dx}{dt} = 3\alpha t^2 + \beta.$$

$$a = \frac{dv}{dt} = 6\alpha t.$$

Force $F = ma = 6mat$.

Instantaneous power:

$$P = Fv = (6mat)(3\alpha t^2 + \beta) \propto (3\alpha t^2 + \beta)t$$

Sol 12. Option (A)

$$P = Fv = \left(mv \frac{dv}{ds}\right) v = mv^2 \frac{dv}{ds}.$$

$$v^2 dv = \frac{P}{m} ds \Rightarrow \int_0^v v^2 dv = \frac{P}{m} \int_0^s ds$$

$$\frac{v^3}{3} = \frac{Ps}{m} \Rightarrow v = \left(\frac{3Ps}{m}\right)^{1/3}$$

Sol 13. Option (A)

From Solution 12: $v = \left(\frac{3Ps}{m}\right)^{1/3} \propto x^{1/3}$.

$$a = v \frac{dv}{dx} \propto (x^{1/3}) \left(\frac{d}{dx} x^{1/3}\right) = (x^{1/3}) \left(\frac{1}{3} x^{-2/3}\right) \propto x^{-1/3}$$

Sol 14. Option (B)

Tension force:

$$T = M(g + a) = 500(10 + 2) = 6000 \text{ N}$$

Speed at $t = 3$ s:

$$v = at = 2 \times 3 = 6 \text{ m/s}$$

Power:

$$P = Tv = 6000 \times 6 = 36000 \text{ W} = 36 \text{ kW}$$

Sol 15. Option (A)

Total force required:

$$F = Mg + f = 1800(10) + 4000 = 18000 + 4000 = 22000 \text{ N}$$

Power in Watts:

$$P = Fv = 22000 \times 3 = 66000 \text{ W}$$

Power in horsepower:

$$P_{\text{hp}} = \frac{66000}{746} \approx 88.47 \text{ hp} \approx 88.5 \text{ hp}$$

Sol 16. Option (A)

Total kinetic energy of N bullets per second:

$$P = N \left(\frac{1}{2} mu^2\right) = \frac{1}{2} Nmu^2$$

Sol 17. Option (A)

Work done in $t = 2$ s:

$$W = \int_0^2 P(t) dt = \int_0^2 3t^2 dt = [t^3]_0^2 = 8 \text{ J}$$

By Work-Energy Theorem:

$$\frac{1}{2} mv^2 = W \implies \frac{1}{2} (2)v^2 = 8 \implies v^2 = 8 \implies v = 2\sqrt{2} \text{ m/s}$$

Sol 18. Option (C)

Mass of air arriving per unit time: $\frac{dm}{dt} = \rho Av$.

Kinetic power of the wind stream:

$$P = \frac{1}{2} \left(\frac{dm}{dt}\right) v^2 = \frac{1}{2} \rho Av^3$$

Sol 19. Option (D)

In uniform circular motion, the net centripetal force is perpendicular to the velocity vector at all times:

$$P = \vec{F}_c \cdot \vec{v} = 0$$

Sol 20. Option (B)

To maintain constant speed, the engine must supply force

$$F = F_{\text{drag}} = cv^2.$$

Power required:

$$P = Fv = (cv^2)v = cv^3 \propto v^3$$